

- Autotrophs, **188f**, 581t, 586, 889, 894f, 1240f
 Auxin, 845, 846t, 847f, **848f**, 849f, 850, 854f, 861
 Avery, Mary Ellen, 943, 944f
 Avery, Oswald, 315
 Avian flu, 410, 1234, 1235f
 Avogadro's number, 50
 Axel, Richard, 1124
 Axillary buds, **761**
 Axis formation, 385f–386f, 387, 1059, 1060f
 Axolotls, 544, 545f, 921f
 Axons, 879f, 1067, **1068f**, 1076f–1077f, 1084, 1086–1087, 1089f, 1090
 Azidothymidine (AZT), 409
Azolla, 813f, 817
- B**
Bacillus anthracis, 575f, 584f
Bacillus thuringiensis, 837, 838f
 Backbone, nucleic acid, 86f
 Backbone, polypeptide, 78f
 Bacteria, **12f**
 alcohol fermentation by, 180f
 anaerobic respiration in, 179–180
 antibiotic drugs and, 349
 antibiotic resistance in, 478f, 581
 binary fission in, 242, 243f
 bioremediation using, 1255f
 Cas9 protein of, 360, 361f
 cell signaling in, 213f, 214
 cell structure of, 97f
 cellular integration and, 121f
 cholera and, 224
 conjugation in, 579, 580f, 581
 as decomposers, 1240f
 in digestive systems, 912, 913f–914f
 diversity of, 1223f
 as DNA cloning vectors, 418f, 419
 DNA packing in chromosomes of, 330–331
 DNA replication in, 322f–327f
 domain, 12f, 568, 569f, 583f–584f, 585t
 evidence for DNA in research on, 315f–316f, 317
 evolution of cell division in, 242, 243f
 evolution of glycolysis in, 182
 expressing cloned eukaryotic genes in, 422–423
 flagellum, movement of, 993f
 genome sizes and number of genes for, 448t, 449
 G proteins and infections by, 218f
 Gram staining of, 574, 575f
 immune system recognition of, 952f
 land colonization by, 536
 as model organism. *See Escherichia coli*
 mutualistic and pathogenic, 587, 588f, 813f
 nitrogen-fixing, and plants, 814f–816f, 817, 1244
 nutrient limitations and, 1244
 origin of photosynthesis in, 533, 534f
 origins of mitochondria and chloroplasts, 109, 110f
 in Permian mass extinction, 540
 photosynthesis of, 188f, 189, 199
 phylogeny of, 583f–584f, 585
 plant defenses against, 866
 in polymerase chain reaction, 421, 422f
 prokaryotic cells of, 6f, 97. *See also* Prokaryotes;
 Prokaryotic cells
 regulation of transcription in, 365f–369f, 370
 relatedness of, to mitochondria, 595
 reproduction rate of, 592
 root mutualism with, 592
 soil, 807
 transcription and translation in, 339f, 355f
 transcription in, 342, 343f–344f
 translation in. *See* Translation
 viral infections of. *See* Phages
 Bacteriophages (phages), **315, 401**
 capsids of, 400f, 401
 Cas9 protein and, 360, 361f
 defense against, 404f
 in DNA research, 315f–316f, 317
 in fighting infections, 872, 931
 prophages and temperate, 402, 403f
 replicative cycles of, 402f–404f
 in transduction, 579f
 virulent, 402f
 Bacteriorhodopsin, 129f
Bacteroides thetaiotaomicron, 587
 Bacteroids, **816f**, 817
 Bada, Jeffrey, 58
- Bait-and-switch defense, 600–601
 Baker, C. S., 557f
 Baker's yeast. *See Saccharomyces cerevisiae*
 Balance, body, 1112f–1116f, 1135
 Balancer, 1066
 Balancing selection, **500**, 501f–503f
Balanus balanoides, 1216f
 Baleen, 903f
 Ball-and-socket joints, 1134f
 Ball-and-stick models, 39f, 59f
 Ballooning, spider, 708
 Ball pythons, 892
 Banana, 268, 313
 Banana slugs, 1155, 1156f
 Barbiturates, 104–105
 Barbs, 645f
 Bar graphs in Scientific Skills Exercises, 23, 179, 376, 483, 590, 629, 700, 762, 864, 1150, 1186, 1217
 Bark, **774**
 Bark beetles, 1244f, 1245, 1280f
 Barley, 650f, 851f
 Barnacles, 708–709, 710f, 1216f
 Barr, Murray, 300
 Barr body, **300f**, 309
 Barrier contraceptives, 1038f
 Barrier defenses, 953–954
 Barrier reefs, 1182f
 Basal angiosperms, **649**, 650f
 Basal animals, 682, 683f, 684, 690
 Basal body, **115**, 116f
 Basal cells, 828f
 Basal lamina, 929f
 Basal-like breast cancer, 393f
 Basal metabolic rate (BMR), **890f**–891f, 892
 Basal nuclei, 1093f
 Basal surface, epithelial, 877f
 Basal taxon, 556f, **557**
 Base pairing, DNA and RNA, 86f, 319f–321f
 Bases, **51**
 amino acids as, 76, 77f
 buffers and, 52–53
 hydrogen ions, acids, and, 51
 pH scale and, 51, 52f
 Basidiocarps, **666f**
 Basidiomycetes, 657, 660f, **665f**–667f
 Basidiospores, 666f
 Basidium, **665**
 Basilar membrane, 1113f
 Basin wetlands, 1179f
 Basking, reptilian, 735–736
 Basophils, 934f–935f
 Bass fishes, 881f
 Bassham, James, 191
 Batesian mimicry, 1218f, **1219**, 1237
Batrachochytrium dendrobatidis, 669, 670f
 Bats, 15f, 717, 825f, 991f, 992, 1262f
 Bautista, Diana, 92
 B cells, **958f**–960, 961f–966f
 DNA of, 976
 Bdelloid rotifers, 266, 267f, 698
 Beach mouse (*Peromyscus polionotus*), 2f, 20f–21f
 Beadle, George, 336f–338f
Beagle, HMS, C. Darwin's voyage on, 471, 472f
 Beaks
 finch, 473f, 486f–487f
 shapes of bird, 740f
 soapberry bug, 477f, 478
 Beans, 829f–830f, 858f, 899
 Bears, 145f, 510f, 1268f, 1269
 Beavers, 1226f
 Bed bugs, 712f
 Beech trees, 1170f
 Bees, 298f, 712f, 824f, 887
 Beetles, 475f, 712f, 717, 1244f, 1245, 1259, 1280f
 Behavior, **1140**. *See also* Animal behavior
 Behavioral ecology, **1140**
 Behavioral isolation, 508f
 Beijerinck, Martinus, 399
 Belding's ground squirrels, 1156, 1158, 1159f, 1193t, 1194f, 1195, 1213
 Beluga whales, 1111f
 Benign tumors, **249**
 Bennettsittes, 648f
 Benson, Andrew, 191
 Benthic zone, **1177f**
 Benthos, **1177**
 Bergmann, Christian, 897
- Berthold, Peter, 1157f
 β_2 -adrenergic receptor, 217f, 220
 β -amyloid, 1104
 β chain, 959f
 β -galactosidase, 159f, 368f
 β -globin, 87, 89, 357f, 358, 364
 β -globin gene family, 453f
 β -keratin, bird feathers and, 738f, 739
 β pleated sheet, **80f**
 Beta-carotene, 838f
 Beta cells, 916f
 Beta oxidation, **183**
 BGI (formerly Beijing Genome Institute), 444
bicoid gene, 386f–387f, 463
 Biennials, 766
 Big-bang reproduction, 1200
 Bilateral symmetry
 animal, 679, 680f, 683
 axis formation and, 1059, 1060f
 flower, 644f, 649f
 Bilaterians, **677**
 chordates, 719f
 Deuterostomia, 713
 Ecdysozoans, 705
 invertebrates, 686f
 Lophotrochozoans, 694
 origin of, 677
 phylogeny of, 683f, 717
 Bilayers, phospholipid. *See* Phospholipid bilayers
 Bile, **909**, 915f
 Binary fission, **242**, 243f, 577–578, 606, 607f
 Binding sites, ribosome, 350f
 Binomials (nomenclature), 470, **554**, 555f
 Biochemical pathways. *See* Metabolic pathways
 Biochemistry, 96
 Biodiversity
 angiosperm, 649f–650f. *See also* Angiosperms
 animal. *See* Animals
 of bacteria and archaea. *See* Archaea; Bacteria
 cellular structures and, 125
 classification of, 470
 climate change effects on, 1279
 conservation biology and, 1261. *See also*
 Conservation biology
 current crisis of, 1260f–1262f
 effects of mass extinctions on, 540f–542f
 evolution and, 14f–16f, 473. *See also* Evolution
 of fungi. *See* Fungi
 gymnosperm, 641, 642f–643f
 habitat loss and fragmentation and, 1263, 1264f
 human welfare and, 1262, 1263f
 invertebrate. *See* Invertebrates
 landscape ecology and regional conservation to
 sustain, 1270, 1271f–1274f
 levels of, 1261f–1262f
 plant. *See* Plants
 protection of, 1260
 protist. *See* Protists
 resurrection of, 1266
 within species, 507f
 sustainable development and, 1284, 1285f, 1286
 taxonomy and classification of, 12f–13f
 threats to, 1260f, 1261, 1263, 1264f–1266f
 tree of life and, 15f–16f. *See also* Phylogenetic
 trees; Phylogenies; Tree of life
 tropical deforestation as threat to, 651f, 652
 unity in, 13f–14f, 26, 469, 473
 vertebrate. *See* Vertebrates
 Biodiversity hot spots, **1272f**
 Bioenergetics, **144, 889**. *See also* Metabolism
 energy allocation and use in, 889, 890f
 energy budgets in, 892
 energy costs of foraging in, 1149
 influences on metabolic rate in, 891f, 892
 of locomotion, 1136
 metabolic rates and thermoregulation in, 890–891
 of osmoregulation, 980
 principles of, 163
 thyroid regulation of, 1009
 torpor, hibernation, and energy conservation in,
 892, 893f
 of urea and uric acid wastes, 983
 Biofilms, 213, **582f**
 Biofortification, 838f
 Biofuels, 671f, **838**
 Biogenic amines, 1081t